ExamGOAL
PYQ PDF

Free · With answer key

JEE MAIN · QUESTION PAPER

JEE Main 2023 (Online) 29th January Morning Shift

JEE Main Question Paper with Answer Key

29 Jan 2023	90	17	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	21 questions
Mathematics	27 questions
Physics	25 questions
Out of syllabus	17 questions
Answer key	at the end

Practise online

Every question with its full solution, free on **questions.examside.com**

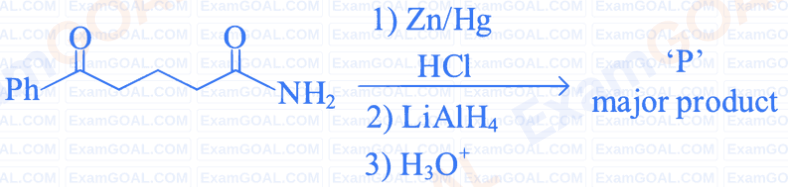
1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

Chemistry 21 questions

Q1. MCQ (Single Correct Answer)

Open solution

The major product 'P' for the following sequence of reactions is :



(A)



(B)



(C)



(D)



Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q2. MCQ (Single Correct Answer)

The magnetic behaviour of Li_2O, Na_2O_2 and KO_2, respectively, are :

(A) paramagnetic, paramagnetic and diamagnetic

(B) paramagnetic, diamagnetic and paramagnetic

(C) diamagnetic, paramagnetic and diamagnetic

(D) diamagnetic, diamagnetic and paramagnetic

Open solution

Q3. MCQ (Single Correct Answer)

Compound that will give positive Lassaigne's test for both nitrogen and halogen is :

(A) $NH_2OH.HCl$ (B) $N_2H_4.HCl$ (C) $CH_3NH_2.HCl$ (D) NH_4Cl

Open solution

Q4. MCQ (Single Correct Answer)

Identify the correct order for the given property for following compounds.

(A) Boiling Point: 

(B) Density: 

(C) Boiling Point: 

(D) Density: 

(E) Boiling Point: 

Choose the correct answer from the option given below :

(A) (A), (B) and (E) only

(B) (A), (C) and (D) only

(C) (A), (C) and (E) only

(D) (B), (C) and (D) only

Open solution

Q5. MCQ (Single Correct Answer)

The bond dissociation energy is highest for

(A) Cl_2 (B) I_2 (C) Br_2 (D) F_2

Open solution

Q6. MCQ (Single Correct Answer)

Number of cyclic tripeptides formed with 2 amino acids A and B is :

(A) 2 (B) 3 (C) 5 (D) 4

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q7. MCQ (Single Correct Answer)

The shortest wavelength of hydrogen atom in Lyman series is λ. The longest wavelength in Balmer series of He^+ is

(A) $\frac{36\lambda}{5}$ (B) $\frac{5}{9\lambda}$ (C) $\frac{9\lambda}{5}$ (D) $\frac{5\lambda}{9}$

Open solution

Q8. MCQ (Single Correct Answer)

Match List I with List II

	List I		List II
	Reaction		Reagents
(A)	Hofmann Degradation	(I)	$\text{Conc. KOH, } \Delta$
(B)	Clemmensen reduction	(II)	$CHCl_3, NaOH/H_3O^{\oplus}$
(C)	Cannizzaro reaction	(III)	$Br_2, NaOH$
(D)	Reimer-Tiemann Reaction	(IV)	$Zn - Hg/HCl$

Choose the correct answer from the options given below :

(A) (A) - III, (B) - IV, (C) - I, (D) - II

(B) (A) - II, (B) - IV, (C) - I, (D) - III

(C) (A) - III, (B) - IV, (C) - II, (D) - I

(D) (A) - II, (B) - I, (C) - III, (D) - IV

Open solution

Q9. MCQ (Single Correct Answer)

The increasing order of pK_a for the following phenols is

(A) 2, 4 - Dinitrophenol

(B) 4 - Nitrophenol

(C) 2, 4, 5 - Trimethylphenol

(D) Phenol

(E) 3-Chlorophenol

Choose the correct answer from the option given below :

(A) (A), (E), (B), (D), (C)

(B) (C), (E), (D), (B), (A)

(C) (C), (D), (E), (B), (A)

(D) (A), (B), (E), (D), (C)

Open solution

Q10. MCQ (Single Correct Answer)

Chiral complex from the following is :

Here en = ethylene diamine

(A) $\text{cis} - [PtCl_2(en)_2]^{2+}$

(B) $\text{trans} - [Co(NH_3)_4Cl_2]^{+}$

(C) $\text{trans} - [PtCl_2(en)_2]^{2+}$

(D) $\text{cis} - [PtCl_2(NH_3)_2]$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q11. MCQ (Single Correct Answer)

The standard electrode potential (M^{3+}/M^{2+}) for V, Cr, Mn & Co are -0.26 V, -0.41 V, + 1.57 V and + 1.97 V, respectively. The metal ions which can liberate H_2 from a dilute acid are :

(A) Mn^{2+} and Co^{2+}

(B) V^{2+} and Mn^{2+}

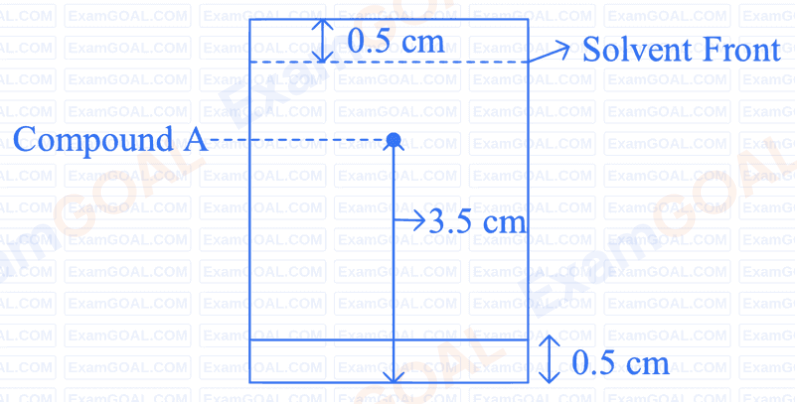
(C) V^{2+} and Cr^{2+}

(D) Cr^{2+} and Co^{2+}

Open solution

Q12. Numerical

Following chromatogram was developed by adsorption of compound 'A' on a 6 cm TLC glass plate. Retardation factor of the compound 'A' is _____ $\times 10^{-1}$.



Open solution

Q13. Numerical

Millimoles of calcium hydroxide required to produce 100 mL of the aqueous solution of pH 12 is $x \times 10^{-1}$. The value of x is _____ (Nearest integer).

Assume complete dissociation.

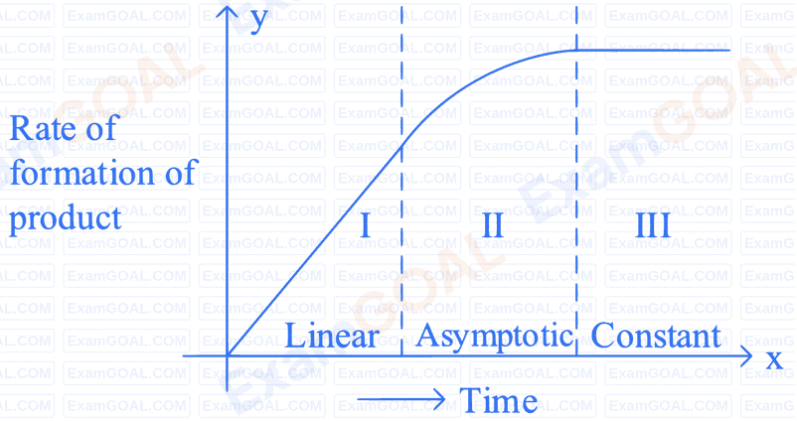
Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q14. Numerical

For certain chemical reaction $X \rightarrow Y$, the rate of formation of product is plotted against the time as shown in the figure. The number of correct statement/s from the following is _____.

Open solution



(A) Over all order of this reaction is one.

(B) Order of this reaction can't be determined.

(C) In region I and III, the reaction is of first and zero order respectively.

(D) In region-II, the reaction is of first order.

(E) In region-II, the order of reaction is in the range of 0.1 to 0.9.

Q15. Numerical

The number of molecules or ions from the following, which do not have odd number of electrons are _____.

Open solution

(A) NO_2

(B) ICl_4^-

(C) BrF_3

(D) ClO_2

(E) NO_2^+

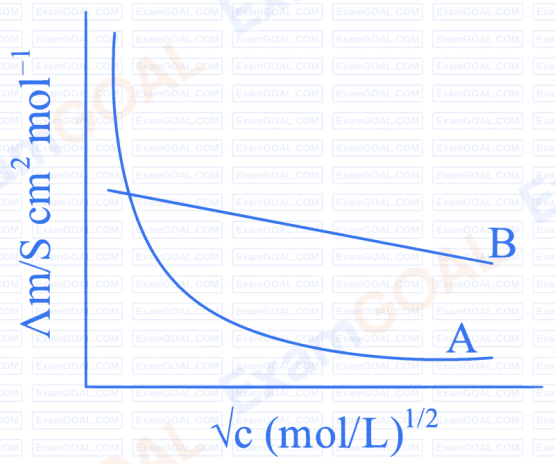
(F) NO

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q16. Numerical

Following figure shows dependence of molar conductance of two electrolytes on concentration. Λ_m^o is the limiting molar conductivity.

Open solution



The number of <u>incorrect</u> statement(s) from the following is _____

(A) Λ_m^o for electrolyte A is obtained by extrapolation

(B) For electrolyte B, Λm vs $\sqrt{c}$ graph is a straight line with intercept equal to Λ_m^o

(C) At infinite dilution, the value of degree of dissociation approaches zero for electrolyte B.

(D) Λ_m^o for any electrolyte A and B can be calculated using λ° for individual ions

Q17. Numerical

Solid Lead nitrate is dissolved in 1 litre of water. The solution was found to boil at 100.15°C. When 0.2 mol of NaCl is added to the resulting solution, it was observed that the solution froze at -0.8° C. The solubility product of $PbCl_2$ formed is _____ $\times 10^{-6}$ at 298 K. (Nearest integer)

Given : $K_b = 0.5$ K kg mol^{-1} and $K_f = 1.8$ K kg mol^{-1}. Assume molality to be equal to molarity in all cases.

Open solution

Q18. Numerical

Water decomposes at 2300 K

$H_2O(g) \rightarrow H_2(g) + \frac{1}{2}O_2(g)$

The percent of water decomposing at 2300 K and 1 bar is _____ (Nearest integer).

Equilibrium constant for the reaction is 2×10^{-3} at 2300 K.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q19. Numerical

Consider the following reaction approaching equilibrium at 27°C and 1 atm pressure

$$A + B \underset{k_r=10^2}{\overset{k_f=10^3}{\rightleftharpoons}} C + D$$

The standard Gibb's energy change $(\Delta_r G^\theta)$ at 27°C is $(-)$ _____ kJ mol^{-1} (Nearest integer).

(Given : $R = 8.3\ J\ K^{-1}\ mol^{-1}$ and $\ln 10 = 2.3$)

Open solution

Q20. Numerical

The sum of bridging carbonyls in $W(CO)_6$ and $Mn_2(CO)_{10}$ is _____.

Open solution

Q21. Numerical

17 mg of a hydrocarbon (M.F. $C_{10}H_{16}$) takes up 8.40 mL of the H_2 gas measured at 0°C and 760 mm of Hg. Ozonolysis of the same hydrocarbon yields



The number of double bond/s present in the hydrocarbon is _____.

Open solution

Mathematics 27 questions

Q22. MCQ (Single Correct Answer)

Let $\lambda \neq 0$ be a real number. Let α, β be the roots of the equation $14x^2 - 31x + 3\lambda = 0$ and α, γ be the roots of the equation $35x^2 - 53x + 4\lambda = 0$. Then $\frac{3\alpha}{\beta}$ and $\frac{4\alpha}{\gamma}$ are the roots of the equation

(A) $7x^2 - 245x + 250 = 0$

(B) $49x^2 - 245x + 250 = 0$

(C) $49x^2 + 245x + 250 = 0$

(D) $7x^2 + 245x - 250 = 0$

Open solution

Q23. MCQ (Single Correct Answer)

Let B and C be the two points on the line $y + x = 0$ such that B and C are symmetric with respect to the origin. Suppose A is a point on $y - 2x = 2$ such that $\triangle ABC$ is an equilateral triangle. Then, the area of the $\triangle ABC$ is :

(A) $\frac{10}{\sqrt{3}}$

(B) $2\sqrt{3}$

(C) $3\sqrt{3}$

(D) $\frac{8}{\sqrt{3}}$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q24. MCQ (Single Correct Answer)

Three rotten apples are mixed accidently with seven good apples and four apples are drawn one by one without replacement. Let the random variable X denote the number of rotten apples. If μ and σ^2 represent mean and variance of X, respectively, then $10(\mu^2+\sigma^2)$ is equal to :

Open solution

(A) 20 (B) 30 (C) 250 (D) 25

Q25. MCQ (Single Correct Answer)

Let $f(\theta)=3\left(\sin^4\left(\frac{3\pi}{2}-\theta\right)+\sin^4(3\pi+\theta)\right)-2(1-\sin^2 2\theta)$ and $S=\left\{\theta\in[0,\pi]: f'(\theta)=-\frac{\sqrt{3}}{2}\right\}$. If $4\beta=\sum\limits_{\theta\in S}\theta$, then $f(\beta)$ is equal to

Open solution

(A) $\frac{9}{8}$ (B) $\frac{3}{2}$ (C) $\frac{5}{4}$ (D) $\frac{11}{8}$

Q26. MCQ (Single Correct Answer)

Fifteen football players of a club-team are given 15 T-shirts with their names written on the backside. If the players pick up the T-shirts randomly, then the probability that at least 3 players pick the correct T-shirt is :

Open solution

(A) $\frac{1}{6}$ (B) $\frac{2}{15}$ (C) $\frac{5}{24}$ (D) 0.08

Q27. MCQ (Single Correct Answer)

Let the tangents at the points $A(4,-11)$ and $B(8,-5)$ on the circle $x^2+y^2-3x+10y-15=0$, intersect at the point C. Then the radius of the circle, whose centre is C and the line joining A and B is its tangent, is equal to :

Open solution

(A) $\frac{2\sqrt{13}}{3}$ (B) $\frac{3\sqrt{3}}{4}$ (C) $\sqrt{13}$ (D) $2\sqrt{13}$

Q28. MCQ (Single Correct Answer)

Let $f(x)=x+\frac{a}{\pi^2-4}\sin x+\frac{b}{\pi^2-4}\cos x, x\in R$ be a function which

satisfies $f(x)=x+\int\limits_0^{\pi/2}\sin(x+y)f(y)dy$. then $(a+b)$ is equal to

Open solution

(A) $-2\pi(\pi+2)$ (B) $-\pi(\pi-2)$ (C) $-\pi(\pi+2)$ (D) $-2\pi(\pi-2)$

Q29. MCQ (Single Correct Answer)

Let α and β be real numbers. Consider a 3×3 matrix A such that $A^2=3A+\alpha I$. If $A^4=21A+\beta I$, then

Open solution

(A) $\alpha=1$ (B) $\alpha=4$ (C) $\beta=8$ (D) $\beta=-8$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q30. MCQ (Single Correct Answer)

A light ray emits from the origin making an angle 30° with the positive x-axis. After getting reflected by the line $x + y = 1$, if this ray intersects x-axis at Q, then the abscissa of Q is :

(A) $\frac{2}{\left(\sqrt{3}-1\right)}$ (B) $\frac{2}{3-\sqrt{3}}$

(C) $\frac{\sqrt{3}}{2\left(\sqrt{3}+1\right)}$ (D) $\frac{2}{3+\sqrt{3}}$

Open solution

Q31. MCQ (Single Correct Answer)

For two non-zero complex numbers z_1 and z_2, if $\mathrm{Re}\,(z_1 z_2) = 0$ and $\mathrm{Re}\,(z_1 + z_2) = 0$, then which of the following are possible?

A. $\mathrm{Im}\,(z_1) > 0$ and $\mathrm{Im}\,(z_2) > 0$

B. $\mathrm{Im}\,(z_1) < 0$ and $\mathrm{Im}\,(z_2) > 0$

C. $\mathrm{Im}\,(z_1) > 0$ and $\mathrm{Im}\,(z_2) < 0$

D. $\mathrm{Im}\,(z_1) < 0$ and $\mathrm{Im}\,(z_2) < 0$

Choose the correct answer from the options given below :

(A) A and C (B) A and B (C) B and D (D) B and C

Open solution

Q32. MCQ (Single Correct Answer)

Let $y = f(x)$ be the solution of the differential equation $y(x+1)dx - x^2 dy = 0, y(1) = e$. Then $\lim\limits_{x \to 0^+} f(x)$ is equal to

(A) e^2 (B) 0 (C) $\frac{1}{e^2}$ (D) $\frac{1}{e}$

Open solution

Q33. MCQ (Single Correct Answer)

Let Δ be the area of the region $\left\{(x, y) \in R^2 : x^2 + y^2 \leq 21, y^2 \leq 4x, x \geq 1\right\}$. Then $\frac{1}{2}\left(\Delta - 21\sin^{-1}\frac{2}{\sqrt{7}}\right)$ is equal to

(A) $2\sqrt{3} - \frac{1}{3}$ (B) $2\sqrt{3} - \frac{2}{3}$ (C) $\sqrt{3} - \frac{4}{3}$ (D) $\sqrt{3} - \frac{2}{3}$

Open solution

Q34. MCQ (Single Correct Answer)

The domain of $f(x) = \frac{\log_{(x+1)}(x-2)}{e^{2\log_e e^x} - (2x+3)}, x \in \mathbb{R}$ is

(A) $(-1, \infty) - \{3\}$ (B) $\mathbb{R} - \{-1, 3\}$ (C) $(2, \infty) - \{3\}$ (D) $\mathbb{R} - \{3\}$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q35. MCQ (Single Correct Answer)

Let $f : R \to R$ be a function such that $f(x) = \frac{x^2+2x+1}{x^2+1}$. Then

(A) $f(x)$ is many-one in $(-\infty, -1)$

(B) $f(x)$ is one-one in $(-\infty, \infty)$

(C) $f(x)$ is one-one in $[1, \infty)$ but not in $(-\infty, \infty)$

(D) $f(x)$ is many-one in $(1, \infty)$

Open solution

Q36. MCQ (Single Correct Answer)

Let $[x]$ denote the greatest integer $\leq x$. Consider the function $f(x) = \max\{x^2, 1 + [x]\}$. Then the value of the integral $\int_0^2 f(x)dx$ is

(A) $\frac{5+4\sqrt{2}}{3}$ (B) $\frac{4+5\sqrt{2}}{3}$ (C) $\frac{8+4\sqrt{2}}{3}$ (D) $\frac{1+5\sqrt{2}}{3}$

Open solution

Q37. MCQ (Single Correct Answer)

Let $A = \left\{(x, y) \in \mathbb{R}^2 : y \geq 0, 2x \leq y \leq \sqrt{4 - (x-1)^2}\right\}$ and

$B = \left\{(x, y) \in \mathbb{R} \times \mathbb{R} : 0 \leq y \leq \min\left\{2x, \sqrt{4 - (x-1)^2}\right\}\right\}$. .

Then the ratio of the area of A to the area of B is

(A) $\frac{\pi}{\pi+1}$ (B) $\frac{\pi-1}{\pi+1}$ (C) $\frac{\pi}{\pi-1}$ (D) $\frac{\pi+1}{\pi-1}$

Open solution

Q38. MCQ (Single Correct Answer)

Consider the following system of equations

$\alpha x + 2y + z = 1$

$2\alpha x + 3y + z = 1$

$3x + \alpha y + 2z = \beta$

for some $\alpha, \beta \in \mathbb{R}$. Then which of the following is NOT correct.

(A) It has a solution for all $\alpha \neq -1$ and $\beta = 2$

(B) It has no solution if $\alpha = -1$ and $\beta \neq 2$

(C) It has no solution for $\alpha = -1$ and for all $\beta \in \mathbb{R}$

(D) It has no solution for $\alpha = 3$ and for all $\beta \neq 2$

Open solution

Q39. MCQ (Single Correct Answer)

If the vectors $\vec{a} = \lambda\hat{i} + \mu\hat{j} + 4\hat{k}$, $\vec{b} = -2\hat{i} + 4\hat{j} - 2\hat{k}$ and $\vec{c} = 2\hat{i} + 3\hat{j} + \hat{k}$ are coplanar and the projection of $\vec{a}$ on the vector $\vec{b}$ is $\sqrt{54}$ units, then the sum of all possible values of $\lambda + \mu$ is equal to :

(A) 24 (B) 0 (C) 18 (D) 6

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q40. MCQ (Single Correct Answer)

Let $x = 2$ be a root of the equation $x^2 + px + q = 0$ and $f(x) = \begin{cases} \frac{1-\cos(x^2-4px+q^2+8q+16)}{(x-2p)^4}, & x \neq 2p \\ 0, & x = 2p \end{cases}$

Then $\lim\limits_{x \to 2p^+} [f(x)]$, where $[.]$ denotes greatest integer function, is

(A) 2 (B) 1 (C) 0 (D) -1

Open solution

Q41. Numerical

Let $a_1, a_2, a_3, \ldots$ be a GP of increasing positive numbers. If the product of fourth and sixth terms is 9 and the sum of fifth and seventh terms is 24, then $a_1a_9 + a_2a_4a_9 + a_5 + a_7$ is equal to _____.

Open solution

Q42. Numerical

Suppose f is a function satisfying $f(x + y) = f(x) + f(y)$ for all $x, y \in N$ and $f(1) = \frac{1}{5}$. If $\sum\limits_{n=1}^{m} \frac{f(n)}{n(n+1)(n+2)} = \frac{1}{12}$, then m is equal to _____.

Open solution

Q43. Numerical

Let the coefficients of three consecutive terms in the binomial expansion of $(1 + 2x)^n$ be in the ratio 2 : 5 : 8. Then the coefficient of the term, which is in the middle of those three terms, is _____.

Open solution

Q44. Numerical

If the co-efficient of x^9 in $\left(\alpha x^3 + \frac{1}{\beta x}\right)^{11}$ and the co-efficient of x^{-9} in $\left(\alpha x - \frac{1}{\beta x^3}\right)^{11}$ are equal, then $(\alpha\beta)^2$ is equal to _____.

Open solution

Q45. Numerical

If all the six digit numbers $x_1\,x_2\,x_3\,x_4\,x_5\,x_6$ with $0 < x_1 < x_2 < x_3 < x_4 < x_5 < x_6$ are arranged in the increasing order, then the sum of the digits in the 72nd number is _____.

Open solution

Q46. Numerical

Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function that satisfies the relation $f(x + y) = f(x) + f(y) - 1, \forall x, y \in \mathbb{R}$. If $f'(0) = 2$, then $|f(-2)|$ is equal to _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q47. Numerical

Five digit numbers are formed using the digits 1, 2, 3, 5, 7 with repetitions and are written in descending order with serial numbers. For example, the number 77777 has serial number 1. Then the serial number of 35337 is _____.

Open solution

Q48. Numerical

Let the co-ordinates of one vertex of ΔABC be $A(0, 2, \alpha)$ and the other two vertices lie on the line $\frac{x+\alpha}{5} = \frac{y-1}{2} = \frac{z+4}{3}$. For $\alpha \in \mathbb{Z}$, if the area of ΔABC is 21 sq. units and the line segment BC has length $2\sqrt{21}$ units, then α^2 is equal to _____.

Open solution

Physics 25 questions

Q49. MCQ (Single Correct Answer)

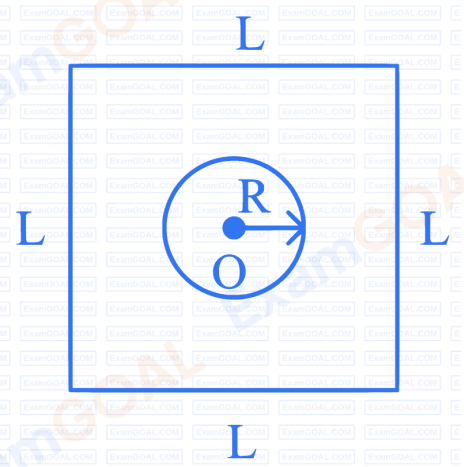
Two particles of equal mass 'm' move in a circle of radius 'r' under the action of their mutual gravitational attraction. The speed of each particle will be :

(A) $\sqrt{\frac{Gm}{4r}}$ (B) $\sqrt{\frac{Gm}{2r}}$ (C) $\sqrt{\frac{Gm}{r}}$ (D) $\sqrt{\frac{4Gm}{r}}$

Open solution

Q50. MCQ (Single Correct Answer)

Find the mutual inductance in the arrangement, when a small circular loop of wire of radius 'R' is placed inside a large square loop of wire of side L ($L \gg R$). The loops are coplanar and their centres coincide :



(A) $M = \frac{\sqrt{2}\mu_0 R}{L^2}$ (B) $M = \frac{2\sqrt{2}\mu_0 R}{L^2}$

(C) $M = \frac{2\sqrt{2}\mu_0 R^2}{L}$ (D) $M = \frac{\sqrt{2}\mu_0 R^2}{L}$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q51. MCQ (Single Correct Answer)

In a cuboid of dimension $2\,\mathrm{L} \times 2\,\mathrm{L} \times \mathrm{L}$, a charge q is placed at the center of the surface 'S' having area of $4\,\mathrm{L}^2$. The flux through the opposite surface to 'S' is given by

(A) $\frac{q}{2\epsilon_0}$ (B) $\frac{q}{3\epsilon_0}$ (C) $\frac{q}{12\epsilon_0}$ (D) $\frac{q}{6\in_0}$

Open solution

Q52. MCQ (Single Correct Answer)

Surface tension of a soap bubble is $2.0 \times 10^{-2}\mathrm{Nm}^{-1}$. Work done to increase the radius of soap bubble from $3.5\ \mathrm{cm}$ to $7\ \mathrm{cm}$ will be:

Take $\left[\pi = \frac{22}{7}\right]$

(A) $18.48 \times 10^{-4}\ \mathrm{J}$ (B) $5.76 \times 10^{-4}\ \mathrm{J}$ (C) $0.72 \times 10^{-4}\ \mathrm{J}$ (D) $9.24 \times 10^{-4}\ \mathrm{J}$

Open solution

Q53. MCQ (Single Correct Answer)

Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: If dQ and dW represent the heat supplied to the system and the work done on the system respectively. Then according to the first law of thermodynamics $dQ = dU - dW$.

Reason R: First law of thermodynamics is based on law of conservation of energy.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both A and R are correct but R is not the correct explanation of A

(B) Both A and R are correct and R is the correct explanation of A

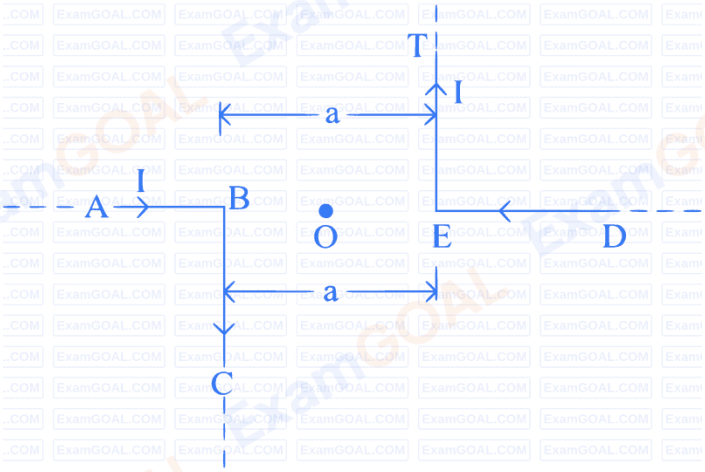
(C) A is correct but R is not correct

(D) A is not correct but R is correct

Open solution

Q54. MCQ (Single Correct Answer)

The magnitude of magnetic induction at mid point O due to current arrangement as shown in Fig will be



(A) $\frac{\mu_0 I}{\pi a}$ (B) $\frac{\mu_0 I}{4\pi a}$ (C) $\frac{\mu_0 I}{2\pi a}$ (D) 0

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q55. MCQ (Single Correct Answer)

A stone is projected at angle 30° to the horizontal. The ratio of kinetic energy of the stone at point of projection to its kinetic energy at the highest point of flight will be -

(A) 1 : 4 (B) 1 : 2 (C) 4 : 3 (D) 4 : 1

Open solution

Q56. MCQ (Single Correct Answer)

A block of mass m slides down the plane inclined at angle 30° with an acceleration $\frac{g}{4}$. The value of coefficient of kinetic friction will be:

(A) $\frac{2\sqrt{3}-1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{1}{2\sqrt{3}}$ (D) $\frac{2\sqrt{3}+1}{2}$

Open solution

Q57. MCQ (Single Correct Answer)

Match List I with List II :

	List I (Physical Quantity)		List II (Dimensional Formula)
A.	Pressure gradient	I.	$\left[\mathrm{M^0L^2\,T^{-2}}\right]$
B.	Energy density	II.	$\left[\mathrm{M^1L^{-1}\,T^{-2}}\right]$
C.	Electric Field	III.	$\left[\mathrm{M^1L^{-2}\,T^{-2}}\right]$
D.	Latent heat	IV.	$\left[\mathrm{M^1\,L^1\,T^{-3}\,A^{-1}}\right]$

Choose the correct answer from the options given below:

(A) A-III, B-II, C-IV, D-I (B) A-III, B-II, C-I, D-IV

(C) A-II, B-III, C-IV, D-I (D) A-II, B-III, C-I, D-IV

Open solution

Q58. MCQ (Single Correct Answer)

A car is moving on a horizontal curved road with radius 50 m. The approximate maximum speed of car will be, if the coefficient of friction between tyres and road is 0.34. [take g = 10 $\mathrm{ms^{-2}}$]

(A) 3.4 $\mathrm{ms^{-1}}$ (B) 13 $\mathrm{ms^{-1}}$ (C) 22.4 $\mathrm{ms^{-1}}$ (D) 17 $\mathrm{ms^{-1}}$

Open solution

Q59. MCQ (Single Correct Answer)

Which of the following are true?

A. Speed of light in vacuum is dependent on the direction of propagation.

B. Speed of light in a medium is independent of the wavelength of light.

C. The speed of light is independent of the motion of the source.

D. The speed of light in a medium is independent of intensity.

Choose the correct answer from the options given below:

(A) B and D only (B) B and C only (C) A and C only (D) C and D only

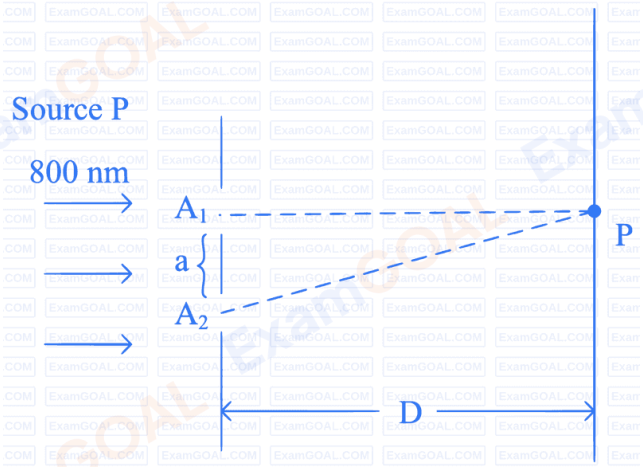
Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q60. MCQ (Single Correct Answer)

Open solution

In a Young's double slit experiment, two slits are illuminated with a light of wavelength $800\ \text{nm}$. The line joining A_1P is perpendicular to A_1A_2 as shown in the figure. If the first minimum is detected at P, the value of slits separation 'a' will be:



The distance of screen from slits D = 5 cm

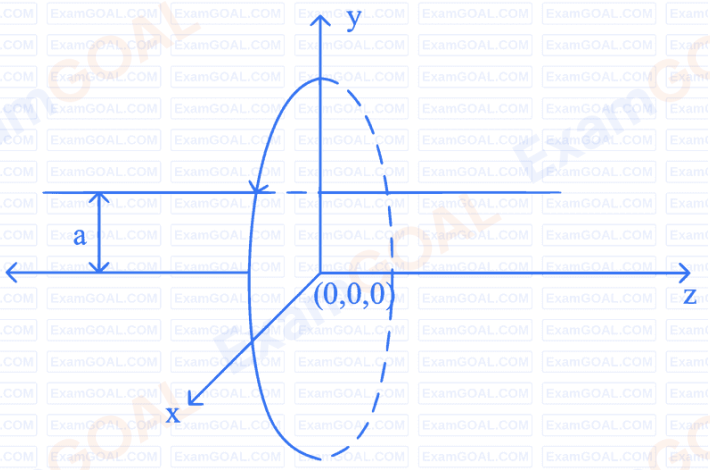
(A) 0.2 mm (B) 0.5 mm (C) 0.4 mm (D) 0.1 mm

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

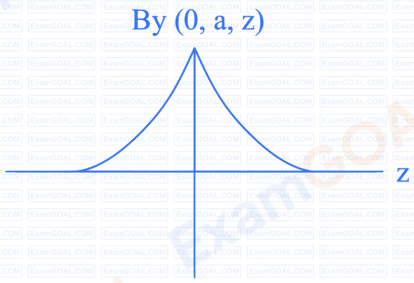
Q61. MCQ (Single Correct Answer)

Open solution

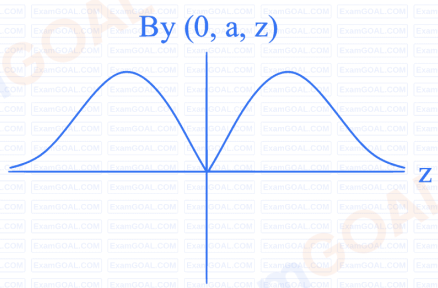
A single current carrying loop of wire carrying current I flowing in anticlockwise direction seen from +ve z direction and lying in xy plane is shown in figure. The plot of $\hat{j}$ component of magnetic field (By) at a distance 'a' (less than radius of the coil) and on yz plane vs z coordinate looks like



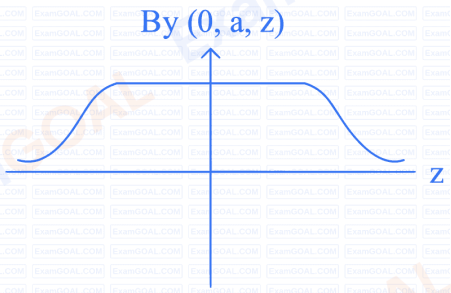
(A)



(B)

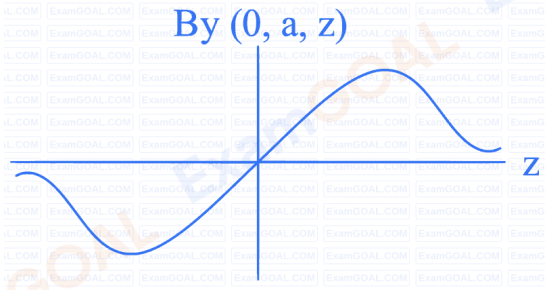


(C)



Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

(D)



Q62. MCQ (Single Correct Answer)

Which one of the following statement is **not** correct in the case of light emitting diodes?

A. It is a heavily doped p-n junction.

B. It emits light only when it is forward biased.

C. It emits light only when it is reverse biased.

D. The energy of the light emitted is equal to or slightly less than the energy gap of the semiconductor used.

Choose the correct answer from the options given below:

(A) A (B) B (C) C and D (D) C

Open solution

Q63. MCQ (Single Correct Answer)

A bicycle tyre is filled with air having pressure of $270\ \mathrm{kPa}$ at $27^{\circ}\mathrm{C}$. The approximate pressure of the air in the tyre when the temperature increases to $36^{\circ}\mathrm{C}$ is

(A) 262 kPa (B) 360 kPa (C) 270 kPa (D) 278 kPa

Open solution

Q64. MCQ (Single Correct Answer)

The threshold wavelength for photoelectric emission from a material is 5500 $\overset{o}{A}$. Photoelectrons will be emitted, when this material is illuminated with monochromatic radiation from a

A. 75 W infra-red lamp

B. 10 W infra-red lamp

C. 75 W ultra-violet lamp

D. 10 W ultra-violet lamp

Choose the correct answer from the options given below :

(A) C only (B) A and D only (C) C and D only (D) B and C only

Open solution

Q65. MCQ (Single Correct Answer)

Ratio of thermal energy released in two resistors R and 3R connected in parallel in an electric circuit is :

(A) 1 : 1 (B) 1 : 27 (C) 1 : 3 (D) 3 : 1

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q66. Numerical

Two simple harmonic waves having equal amplitudes of 8 cm and equal frequency of 10 Hz are moving along the same direction. The resultant amplitude is also 8 cm. The phase difference between the individual waves is _____ degree.

Open solution

Q67. Numerical

A solid sphere of mass 2 kg is making pure rolling on a horizontal surface with kinetic energy 2240 J. The velocity of centre of mass of the sphere will be _____ ms^{-1}.

Open solution

Q68. Numerical

In a metre bridge experiment the balance point is obtained if the gaps are closed by 2Ω and 3Ω. A shunt of X Ω is added to 3Ω resistor to shift the balancing point by 22.5 cm. The value of X is _____.

Open solution

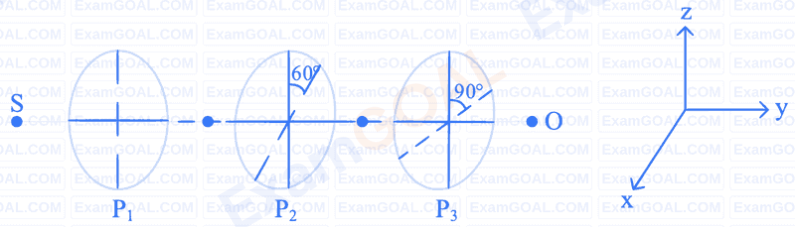
Q69. Numerical

A point charge $q_1 = 4q_0$ is placed at origin. Another point charge $q_2 = -q_0$ is placed at $x = 12$ cm. Charge of proton is q_0. The proton is placed on x axis so that the electrostatic force on the proton is zero. In this situation, the position of the proton from the origin is _____ cm.

Open solution

Q70. Numerical

As shown in the figure, three identical polaroids P_1, P_2 and P_3 are placed one after another. The pass axis of P_2 and P_3 are inclined at angle of 60° and 90° with respect to axis of P_1. The source S has an intensity of 256 $\frac{W}{m^2}$. The intensity of light at point O is _____ $\frac{W}{m^2}$.



Open solution

Q71. Numerical

A 0.4 kg mass takes 8s to reach ground when dropped from a certain height 'P' above surface of earth. The loss of potential energy in the last second of fall is _____ J.

(Take g = 10 m/s^2)

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q72. Numerical

A certain elastic conducting material is stretched into a circular loop. It is placed with its plane perpendicular to a uniform magnetic field B = 0.8 T. When released the radius of the loop starts shrinking at a constant rate of 2 cms^{-1}. The induced emf in the loop at an instant when the radius of the loop is 10 cm will be _____ mV.

Open solution

Q73. Numerical

A tennis ball is dropped on to the floor from a height of 9.8 m. It rebounds to a height 5.0 m. Ball comes in contact with the floor for 0.2s. The average acceleration during contact is _____ ms^{-2}.

(Given g = 10 ms^{-2})

Open solution

Out of syllabus 17 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q74. MCQ (Single Correct Answer) · **Chemistry**

Match List I with List II.

	List I		**List II**
	Antimicrobials		**Names**
(A)	Narrow Spectrum Antibiotic	(I)	Furacin
(B)	Antiseptic	(II)	Sulphur dioxide
(C)	Disinfectants	(III)	Penicillin G
(D)	Broad spectrum antibiotic	(IV)	Chloramphenicol

Choose the correct answer from the options given below :

(A) (A) - III, (B) - I, (C) - II, (D) - IV

(B) (A) - II, (B) - I, (C) - IV, (D) - III

(C) (A) - I, (B) - II, (C) - IV, (D) - III

(D) (A) - III, (B) - I, (C) - IV, (D) - II

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q75. MCQ (Single Correct Answer) · **Chemistry**

The correct order of hydration enthalpies is

(A) K^+

(B) Rb^+

(C) Mg^{2+}

(D) Cs^+

(E) Ca^{2+}

Choose the correct answer from the options given below :

(A) E > C > A > B > D

(B) C > A > E > B > D

(C) C > E > A > D > B

(D) C > E > A > B > D

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

Correct statement about smog is :

(A) NO_2 is present in classical smog

(B) Both NO_2 and SO_2 are present in classical smog

(C) Photochemical smog has high concentration of oxidizing agents

(D) Classical smog also has high concentration of oxidizing agents

Open solution

Q77. MCQ (Single Correct Answer) · **Chemistry**

During the borax bead test with $CuSO_4$, a blue green colour of the bead was observed in oxidising flame due to the formation of :

(A) CuO **(B)** $Cu(BO_2)_2$ **(C)** Cu_3B_2 **(D)** Cu

Open solution

Q78. MCQ (Single Correct Answer) · **Chemistry**

Which of the given compounds can enhance the efficiency of hydrogen storage tank?

(A) SiH_4

(B) Li/P_4

(C) $NaNi_5$

(D) Di-isobutylaluminium hydride

Open solution

Q79. MCQ (Single Correct Answer) · **Chemistry**

"A" obtained by Ostwald's method involving air oxidation of NH_3, upon further air oxidation produces "B", "B" on hydration forms an oxoacid of Nitrogen along with evolution of "A". The oxoacid also produces "A" and gives positive brown ring test.

Identify A and B, respectively :

(A) N_2O_3, NO_2 **(B)** NO_2, N_2O_5 **(C)** NO_2, N_2O_4 **(D)** NO, NO_2

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q80. MCQ (Single Correct Answer) · **Chemistry**

Which of the following salt solutions would coagulate the colloid solution formed when $FeCl_3$ is added to $NaOH$ solution, at the fastest rate?

(A) 10 mL of 0.15 mol dm^{-3} $CaCl_2$

(B) 10 mL of 0.2 mol dm^{-3} $AlCl_3$

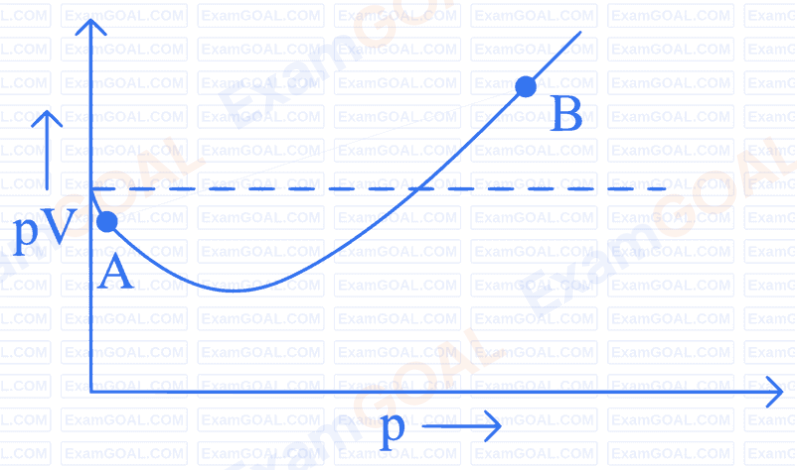
(C) 10 mL of 0.1 mol dm^{-3} Na_2SO_4

(D) 10 mL of 0.1 mol dm^{-3} $Ca_3(PO_4)_2$

Open solution

Q81. MCQ (Single Correct Answer) · **Chemistry**

For 1 mol of gas, the plot of pV vs. p is shown below. p is the pressure and V is the volume of the gas



What is the value of compressibility factor at point A ?

(A) $1+\frac{b}{V}$ (B) $1-\frac{a}{RTV}$ (C) $1+\frac{a}{RTV}$ (D) $1-\frac{b}{V}$

Open solution

Q82. MCQ (Single Correct Answer) · **Chemistry**

The reaction representing the Mond process for metal refining is _____.

(A) $ZnO + C \xrightarrow{\Delta} Zn + CO$

(B) $2K[Au(CN)_2] + Zn \xrightarrow{\Delta} K_2[Zn(CN)_4] + 2Au$

(C) $Zr + 2I_2 \xrightarrow{\Delta} ZrI_4$

(D) $Ni + 4CO \xrightarrow{\Delta} Ni(CO)_4$

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

If p, q and r are three propositions, then which of the following combination of truth values of p, q and r makes the logical expression $\{(p \vee q) \wedge ((\sim p) \vee r)\} \rightarrow ((\sim q) \vee r)$ false?

(A) $p = F, q = T, r = F$

(B) $p = T, q = T, r = F$

(C) $p = T, q = F, r = T$

(D) $p = T, q = F, r = F$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q84. Numerical · **Mathematics**

Let the equation of the plane P containing the line $x+10=\frac{8-y}{2}=z$ be $ax+by+3z=2(a+b)$ and the distance of the plane P from the point (1, 27, 7) be c. Then $a^2+b^2+c^2$ is equal to _____.

Open solution

Q85. Numerical · **Mathematics**

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-zero non-coplanar vectors. Let the position vectors of four points A, B, C and D be $\vec{a}-\vec{b}+\vec{c}, \lambda\vec{a}-3\vec{b}+4\vec{c}, -\vec{a}+2\vec{b}-3\vec{c}$ and $2\vec{a}-4\vec{b}+6\vec{c}$ respectively. If $\overrightarrow{AB}, \overrightarrow{AC}$ and $\overrightarrow{AD}$ are coplanar, then λ is equal to _____.

Open solution

Q86. MCQ (Single Correct Answer) · **Physics**

If a radioactive element having half-life of $30\ \text{min}$ is undergoing beta decay, the fraction of radioactive element remains undecayed after $90\ \text{min}$. will be

(A) $\frac{1}{16}$ (B) $\frac{1}{4}$ (C) $\frac{1}{8}$ (D) $\frac{1}{2}$

Open solution

Q87. MCQ (Single Correct Answer) · **Physics**

A person observes two moving trains, 'A' reaching the station and 'B' leaving the station with equal speed of $30\ \text{m/s}$. If both trains emit sounds with frequency $300\ \text{Hz}$, (Speed of sound: $330\ \text{m/s}$) approximate difference of frequencies heard by the person will be:

(A) 10 Hz (B) 55 Hz (C) 80 Hz (D) 33 Hz

Open solution

Q88. MCQ (Single Correct Answer) · **Physics**

If the height of transmitting and receiving antennas are 80 m each, the maximum line of sight distance will be:

Given : Earth's radius $= 6.4\times10^6$ m

(A) 28 km (B) 64 km (C) 32 km (D) 36 km

Open solution

Q89. Numerical · **Physics**

A body cools from 60°C to 40°C in 6 minutes. If, temperature of surroundings is 10°C. Then, after the next 6 minutes, its temperature will be _____°C.

Open solution

Q90. Numerical · **Physics**

A radioactive element $^{242}_{92}$X emits two α-particles, one electron and two positrons. The product nucleus is represented by $^{234}_{P}$Y. The value of P is _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Answer Key

Chemistry

1. **B**	2. **D**	3. **C**	4. **C**	5. **A**
6. **D**	7. **C**	8. **A**	9. **D**	10. **A**
11. **C**	12. **6**	13. **5**	14. **1**	15. **3**
16. **2**	17. **13**	18. **2**	19. **6**	20. **0**
21. **3**				

Mathematics

22. **B**	23. **D**	24. **A**	25. **C**	26. **D**
27. **A**	28. **A**	29. **D**	30. **D**	31. **D**
32. **B**	33. **C**	34. **C**	35. **C**	36. **A**
37. **B**	38. **C**	39. **A**	40. **C**	41. **60**
42. **10**	43. **1120**	44. **1**	45. **32**	46. **3**
47. **1436**	48. **9**			

Physics

49. **A**	50. **C**	51. **D**	52. **A**	53. **B**
54. **A**	55. **C**	56. **C**	57. **A**	58. **B**
59. **D**	60. **A**	61. **D**	62. **D**	63. **D**
64. **C**	65. **D**	66. **120**	67. **40**	68. **2**
69. **24**	70. **24**	71. **300**	72. **10**	73. **120**

Out of syllabus

74. **A**	75. **D**	76. **C**	77. **B**	78. **C**
79. **D**	80. **B**	81. **B**	82. **D**	83. **A**
84. **355**	85. **2**	86. **C**	87. **B**	88. **B**
89. **28**	90. **87**			

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

ExamGOAL
PYQ PDF

Your Exam. Our Responsibility.

From first chapter to exam day.

Practise with the questions your exam has already asked, then test yourself under real exam conditions.

PAST YEAR QUESTIONS

Every past paper, chapter-wise.

Previous year questions sorted the way you study, each with a full solution.

- Chapter-wise and year-wise
- Weightage and topic tags on every chapter
- Your progress on each one, at a glance

MOCK TESTS

Mock tests with a real exam feel.

Practise under the same pressure as the real paper, on your phone or on the big screen.

- Timed sections, just like exam day
- Mark questions for review and come back

ANALYSIS

Know your score. Know your rank.

A full report after every test, so you know exactly where you stand.

- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

Solve. Check. Understand.

Learn from every question, not just the ones you get right.

- Instant answers as you practise
- Step-by-step explanations
- See how everyone else answered

NCERT Exemplar
With full solutions

Exam syllabus
Subject and chapter-wise

Bookmarks
Organised in groups

Your notes
On any question

ExamGOAL
LEARNING APP

ExamGOAL: Exam Preparation App

One app for every exam. Free trial for new users, on Android and the web.

GET IT ON Google Play

Scan to get the app

www.examgoal.com
Practice and mock tests

questions.examside.com
Free solutions

pdf.questions.examgoal.com
Free PYQ PDFs